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Subject Computer Science Semester Roll No 29616337

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9.01.26

Expt. No. 01 Page No. 1 Date

- Topic of Experiment

Get a lucky winner from a list using random module in Python.

- Hardware Required

CPU core i3 and above, monitor, mouse, keyboard.

- Software required

Python 3.7 or above, IDLE IDE or Sublime Text editor.

- Program

```
import random
names = ['Aakash', 'Subh', 'Nayan', 'Ganpat', 'Nishu']

winner = random.choice (names)

print ("Lucky winner is - ", winner)
```

- Sample Output

Lucky winner is - Ganpat

Teacher's Signature : _____

Topic of Experiment

Arranging three input numbers in ascending order using Python.

Hardware Required

CPU core i3 and above, monitor, mouse, keyboard

Software Required

Python 3.7 or above, IDLE IDE or Sublime Text Editor

Program

```
# User Inputs :
a = int (input ('Enter First no. :'))
b = int (input ('Enter second no. :'))
c = int (input ('Enter Third no. :'))

numbers = [a, b, c]
numbers.sort()

print ("Numbers in ascending order - ")
print (numbers)
```

Output

```
Enter First no. : 15
Enter second no. : 3
Enter Third no. : 9
Numbers in ascending order -
[3, 9, 12]
```

Teacher's Signature : _____

Topic of Experiment

Reading a file's entire content using Python.

Hardware Required

CPU core i3 and above, monitor, mouse, keyboard.

Software Required

Python 3.7 or above, IDLE IDE or Sublime Text Editor

Program

```
fout = open ("poem.txt", 'r')
data = fout.read()
print (data)
fout.close()
```

Output

```
Twinkle, Twinkle, little stars,
How I wonder what you are
Up above the world so high,
Like a diamond in the sky.
```

Teacher's Signature :

Expt. No. 04 Page No. 4 Date

Topic of Experiment

Getting roll no., names and marks of students and storing them in a file named "marks.txt".

Hardware Required

CPU core i3 and above, monitor, mouse, Keyboard

Software Required

Python 3.7 or above, IDLE IDE or sublime Text editor

Program

```
fout = open("marks.txt", 'w')
n = int(input("Enter number of students : "))

for i in range(n):
        roll = input('enter roll no. - ')
        name = input('Enter your name - ')
        marks = input('Enter marks - ')

        fout.write(roll + " " + name + " " + marks + "\n")

fout.close()
print("Data stored successfully")
```

Output

```
Enter number of students : 1
Enter roll no. - 1
Enter your name - xyz
Enter marks - 85
Data stored successfully.
```

Teacher's Signature : _____

Expt. No. 05 Page No. 5 Date

Topic of Experiment

Reading a text file and displaying the count of vowels and consonants using python.

Hardware Required

CPU core i3 or above, IDLE IDE, monitor mouse, keyboard.

Software Required

Python 3.7 or above, VS code editor or sublime text editor.

Program

```
fout = open ("text.txt", 'r')
data = fout.read()
fout.close()

vowels = 0
consonants = 0

for ch in data:
        if ch.isalpha():
                if ch.lower() in "aciou":
                        vowels += 1
                else:
                        consonants += 1
print ("No. of vowels :", vowels)
print ("No. of consonants :", consonants)
```

Teacher's Signature :

Output

No. of vowels : 18
No. of consonants : 32

Teacher's Signature : _____

Expt. No. 6 Page No. 7 Date

Topic of Experiment

Opening an existing file and displaying the student records stored in it using python.

Hardware Required.

CPU core i3 and above, monitor, mouse, keyboard.

Software Required

Python 3.7 or above, IDLE IDE or sublime Text Editor.

Program

```
fout = open ("marks.txt", 'r')
print ("student's records:")

for line in fout:
        print (line)

fout.close()
```

Output

```
Student's records:
1   Aman    85
2   Neha    82
3   Aanchal 96
```

Teacher's Signature :

Topic of Experiment

Updating records of file Stud.dat and displaying students who has scored more than 81 marks.

Hardware Required

CPU core i3 or above, monitor, mouse, keyboard.

Software Required

Python 3.7 or above, IDLE IDE or Sublime text editor

Program

```
fout = open ("stud.dat", 'r')
print ("students scoring more than 81 marks :- ")

for line in fout :
        data = line.split()
        roll = data[0]
        name = data[1]
        marks = int(data[2])
        if marks > 81:
                print (roll, name, marks)
fout.close()
```

Output

```
Students scoring more than 81 marks :
2    Neha    92
4    Rahul   88
```

Teacher's Signature : _____

Expt. No. 08 Page No. 9 Date

Topic of Experiment

Reading and displaying the contents of "employee.csv"

Hardware Required

CPU core i3 and above, monitor, mouse, Keyboard

Software Required

Python 3.7 or above, IDLE IDE or sublime text editor

Program

```
import csv

fout = open("employee.csv", 'r')
reader = csv.reader(fout)

for i in reader:
        print(i)

fout.close()
```

Output

```
['EmpID', 'Name', 'salary']
['101', 'Subh', '93000']
['102', 'Grans', '90001']
['103', 'Nayan', '88008']
```

Teacher's Signature : _____

Expt. No. 09 Page No. 10 Date

Topic of Experiment

Reading and displaying the contact details from the file "employees.csv" using Python.

Hardware Required

CPU core i3 and above, monitor, mouse, keyboard

Software Required

Python 3.7 or above, IDLE IDE or sublime text editor.

Program

```
import csv
fout = open("employees.csv", 'r')
reader = csv.reader(fout)
print("Employee Contacts :")

for row in reader :
        print("Name :", row[1], "Contact", row[3])
fout.close()
```

Output

```
Employee Contacts :

Name : Nayan Contact : 6028372218
Name : Subh Contact : 01125532553
Name : Gans Contact : 9530395303
```

Teacher's Signature :

Expt. No. 10 Page No. 11 Date

Topic of Experiment

Write a program to read details of sports performance of your school and store into a csv file delimited with tab character.

Hardware Required

CPU core i3 and above, keyboard, mouse, moniter.

Software Required

Python 3.7 or above, idle IDE or Sublime text editor.

Program :

```
import csv
fn = open ("sports.csv", 'w')
swritter = csv.writter (fn, deliminater = '\t')
swritter.writerow (['sport', 'competition', 'prizes', 'won'])
ans = 'y'
i = 1

while ans == 'y' :
        print ("Record", i)
        sport = input ("sports name : ")
        comp = int (input ("No. of competetion
                            participated : "))
        prizes = int (input ("Total prizes won :"))
        srec = [sport, comp, prizes]
        swritter.writerow (srec)
        i = i+1
```

Teacher's Signature :

```
        ans = input("want to enter more record ? (y/n):")
fh.close()
```

- Output :

```
Record 1
sport name : Tennis
No. of competition participated : 14
Prize won : 9
want to enter more records ? (y/n) : y

Record 2
sport name : Cricket
No. of competition participated : 93
Prizes won : 49
want to enter more records ? (y/n) : n
```

Topic of Experiment

Getting student's roll number, name and marks from user and storing them in a binary file using python.

Hardware Required

CPU core i3 and above, monitor, mouse, keyboard

Software Required

Python 3.7 or above, IDLE IDE or Sublime Text editor.

Program

```
import pickle
fout = open("student.dat", "wb")
n = int(input("Enter number of students :"))

for i in range(n):
        roll = int(input("Enter roll number :"))
        name = input("Enter name:")
        marks = int(input("Enter marks :"))
        data = [roll, name, marks]
        pickle.dump(data, fout)

fout.close()
print("Student data stored successfully!")
```

Teacher's Signature : _____

Output

```
Enter number of students : 2
Enter roll number : 1
Enter ~~mar~~ name : Subh
Enter marks : 99
Enter roll number : 2
Enter name : Nayan
Enter marks : 92

Student data stored successfully !
```

Teacher's Signature : _____

Topic of Experiment

Implementing stack operations (push and pop) using python.

Hardware Required

CPU core i3 and above, monitor, mouse, keyboard.

Software Required

Python 3.7 or above, IDLE IDE or Sublime Text Editor.

Program

```
stack = []

while True:
    print("\n Stack Operations")
    print ("1. Push")
    print ("2. Pop")
    print ("3. Display")
    print ("4. Exit")
    ch = int(input("Enter your choice:"))

    if ch == 1:
        item = input("Enter element to push:")
        stack.append(item)
        print("Element pushed")
```

Teacher's Signature : _____

```
elif ch == 2:
    if stack == []:
        print("Stack is empty")
    else:
        print("Popped element :", stack.pop())

elif ch == 3:
    print("Stack elements :", stack)

elif ch == 4:
    break

else:
    print("Invalid choice")
```

- Output :-

```
Stack Operations
1. Push
2. Pop
3. Display
4. Exit
Enter your choice : 1
Enter element to push : 10
Element pushed

Enter your choice : 3
Stack elements : ['10']
```

▣ Topic of Experiment
Structure Query language programs

▣ Hardware required :-
CPU core i3 and above, monitor, Keyboard, mouse

▣ Software Required :-
mysql 5.5 and above

Display all the record (all column) from table employee

→ select * from Employee ;

Display Empno. and Ename of all employees from table Employee.

→ select empno , Ename
from Employee ;

Display Ename, sal, and sal added with common from table Employee.

→ select Ename, sal, sal + comm
from Employee ;

Write a query to display employee name, salary and department no. who are not getting commission from table employee.

Teacher's Signature : _____

→ Select Ename, sal, deptno.
from Employee
where comm is Null ;

Write a query to display employee number, name, sal and (sal * 12) as annual salary whose comm is not null from table Employee.

→ select Emp no, ename, sal * 12 "Annual salary" from empl where comm is not null ;

list all the department numberns in the table employee.

→ select dept no. from empl :

list all the unique department numbes in the table employee.

→ Select distinct dept no. from employee ;

list details of all clerks who have not been assigned department is null

→ select * from employee where Job = 'clerk' and dept no is null ;

list the details of the employees whose annual salary is between 25000 - 40000.

Teacher's Signature : _____

→ select * from employee where sal * 12 between 25000 and 40000;

How many job types are offered to employees?

→ select distinct job from empl;

List the details of employees who earn more commission than their salaries.

→ select * from employee where comm > sal;

Write a query to display the names, Job, title and salary of employee who do not have manage.

→ select ename, job, sal, from employee where mgu is null;

Write a query to display the name of employee whose name contains 'A' as third of alphabet.

→ select ename from employee where ename like '__ A %';

Write a query to display the name of employee who is having 'L' as any alphabet of the name.

→ select ename from employee where ename like '% L %';

Teacher's Signature : _____

Expt. No. 14 Page No. 20 Date

Topic of Experiment

Fetching first three records from the student table of MySQL database "test" using python.

Hardware Required

CPU core i3 and above, monitor, mouse, keyboard

Software Required

Python 3.7 or above
MySQL server
MySQL connector for Python
IDLE IDE or any python editor

Program

```
import mysql.connector

mydb = mysql.connector.connect(
        host = "localhost",
        user = "learner",
        password = "fast",
        database = "test")

cursor = mydb.cursor()
cursor.execute("select * from student limit 3")

records = cursor.fetchall()

print("First three records from student table:\n")
```

Teacher's Signature : _____

```
for row in records:
        print(row)

mydb.close()
```

- Output

```
First three records from Student table :
(1, 'Gans', 85)
(2, 'Subh', 92)
(3, 'Nan', 78)
```

Topic of Experiment

Implementing Queue operations (Insertion and Deletion) using Python.

Hardware Required

CPU core i3 and above, monitor, mouse, keyboard.

Software Required

Python 3.7 or above, IDLE IDE or any python editor.

Program

```
queue = []

while True :
    print ("\n QUEUE OPERATIONS")
    print (" 1. Insert")
    print (" 2. Delete")
    print (" 3. Display")
    print (" 4. Exit")

    ch = int(input ("Enter your choice: "))
    item = input ("Enter element to insert:")
    if ch == 1:
        item. append (item)
        print ("Element inserted into queue")
```

Teacher's Signature : _____

```
elif ch == 2:
    if queue == []:
        print ("Queue is empty")
    else :
        print ('Deleted element:',
            queue.pop(0))

elif ch == 3:
    print ("Queue elements :", queue)

elif ch == 4:
    break

else :
    print ("Invalid choice")
```

Output

```
QUEUE OPERATIONS
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice : 1
Enter element to insert : A
Element inserted into queue

Enter your choice : 1
Enter element to insert : B
```

Element inserted into queue.

Enter you choice : 3
Queue elements : ['A', 'B']

Enter your choice : 2
Deleted element : A.